Basic Concepts

- In an optimization problem, the objective is to optimize(maximize or minimize) some function f.
- This function f is called the objective function. It is the focal point or goal of our optimization problem.
- In most optimization problems the objective function f depends on several variables: x_1, x_2,… x_n (called control variables)

Basic Concepts

- Optimization theory develops methods for optimal choices of x_1, x_2,… x_n which maximize (or minimize) the objective function f, that is, methods for finding optimal values of x_1, x_2,… x_n .
- unconstrained optimization
- constrained optimization

The method of steepest descent or gradient method

- The idea of this method is to find a minimum of f(x) by repeatedly computing minima of a function g(t) of a single variable t.
- Suppose that f has a minimum at X_0 and we start at a point x.

$$\mathbf{z}(t) = \mathbf{x} - t\nabla f(\mathbf{x}) \tag{2}$$

$$g(t) = f(\mathbf{z}(t)) \tag{3}$$

z(t) as our next approximation to solution X_0.

EXAMPLE 1

Determine a minimum of

$$f(\mathbf{x}) = x_1^2 + 3x_2^2, \tag{4}$$

starting from $\mathbf{x}_0 = (6, 3) = 6\mathbf{i} + 3\mathbf{j}$ and applying the method of steepest descent.

Solution: page 952

Solution. Clearly, inspection shows that $f(\mathbf{x})$ has a minimum at $\mathbf{0}$. Knowing the solution gives us a better feel of how the method works. We obtain $\nabla f(\mathbf{x}) = 2x_1\mathbf{i} + 6x_2\mathbf{j}$ and from this

$$\mathbf{z}(t) = \mathbf{x} - t\nabla f(\mathbf{x}) = (1 - 2t)x_1\mathbf{i} + (1 - 6t)x_2\mathbf{j}$$

$$g(t) = f(\mathbf{z}(t)) = (1 - 2t)^2x_1^2 + 3(1 - 6t)^2x_2^2.$$

We now calculate the derivative

$$g'(t) = 2(1 - 2t)x_1^2(-2) + 6(1 - 6t)x_2^2(-6),$$

set $g'(t) = 0$, and solve for t, finding

$$t = \frac{x_1^2 + 9x_2^2}{2x_1^2 + 54x_2^2}.$$

Starting from $x_0 = 6\mathbf{i} + 3\mathbf{j}$, we compute the values in Table 22.1, which are shown in Fig. 473.

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right] = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}.$$

Table 22.1 Method of Steepest Descent, Computations in Example 1

n	$\mathbf{x}$		t	$1 - 2t$	$1 - 6t$
0	6.000	3.000	0.210	0.581	−0.258
1	3.484	−0.774	0.310	0.381	−0.857
2	1.327	0.664	0.210	0.581	−0.258
3	0.771	−0.171	0.310	0.381	−0.857
4	0.294	0.147	0.210	0.581	−0.258
5	0.170	−0.038	0.310	0.381	−0.857
6	0.065	0.032			

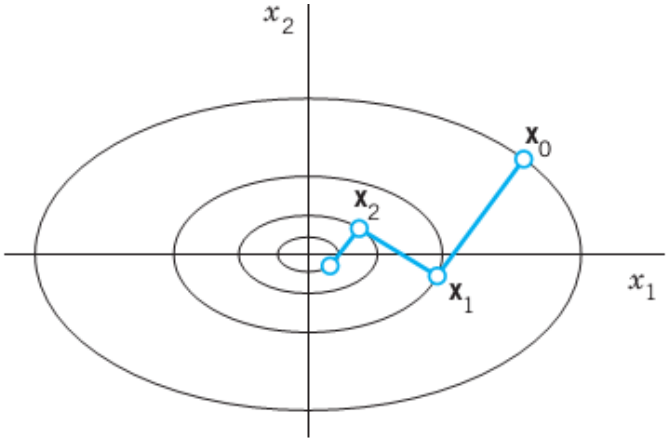


Fig. 473. Method of steepest descent in Example 1

Exercise

- Do 3 steepest descent steps when:

$$f(x) = x_1^2 + 2x_2^2 - x_1 - 6x_2, \ \mathbf{x}_0 = [0 \quad 0]^T$$

Solution

$$f(\mathbf{x}) = (x_1 - 0.5)^2 + 2(x_2 - 1.5)^2 - 4.75$$

$$\nabla f(x) = (2x_1 - 1)\,i + (4x_2 - 6)\,j$$

$$z(t) = x - t\,\nabla f(x)$$

$$z(t) = (x_1 - 2x_1 t + t)\,i + (x_2 - 4x_2 t + 6t)\,j$$

$$g(t) = f(z(t)) = (x_1 - 2x_1 t + t - 0{,}5)^2 + 2(x_2 - 4x_2 t + 6t - 1{,}5)^2 - 4{,}75$$

$$g'(t) = 0 \rightarrow t = \frac{[(2x_1 - 1)(x_1 - 0{,}5) + (4x_2 - 6)(x_2 - 1{,}5)]}{(2x_1 - 1)^2 + (4x_2 - 6)^2}$$

Step	x_1	x_2	$f(\mathbf{x})$
1	0.25342	1.5206	−4.6884
2	0.49351	1.4805	−4.7492
3	0.49680	1.5003	−4.7500

Exercise

- Do 7 steepest descent steps when:

$$f(\mathbf{x}) = x_1^2 + 0.1x_2^2 + 8x_1 + x_2 + 22.5,$$
$$x_0 = [2 \quad -1]^T$$

Solution

$f(\mathbf{x}) = (x_1 + 4)^2 + 0.1(x_2 + 5)^2 + 4$. The computation gives:

Step	x_1	x_2	$f(\mathbf{x})$
1	−4.0240	−1.4016	5.2958
2	−3.7933	−4.8622	4.0449
3	−4.0008	−4.8760	4.0015
4	−3.9929	−4.9952	4.0000
5	−4.0000	−4.9957	4.0000
6	−3.9997	−5.0000	4.0000
7	−4.0000	−5.0000	4.0000

Linear Programming

- Linear programming or linear optimization consists of methods for solving optimization problems with constraints, that is, methods for finding a maximum (or a minimum) $x=(x_1, x_2, \ldots, x_n)$ of a linear objective function

$$z = f(\mathbf{x}) = a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

satisfying the constraints, linear inequalities.

EXAMPLE 1

- Maximize: $$\boxed{z = f(\mathbf{x}) = 40x_1 + 88x_2}$$

constraints

$$
\begin{array}{ll}
(0) & z = 40x_1 + 88x_2 \\
(1) & 2x_1 + 8x_2 \leqq 60 \\
(2) & 5x_1 + 2x_2 \leqq 60 \\
(3) & x_1 \geqq 0 \\
(4) & x_2 \geqq 0
\end{array}
$$

Conversion of Inequalities by the Use of Slack Variables

$$\boxed{\begin{aligned} 2x_1 + 8x_2 &\leqq 60 \\ 5x_1 + 2x_2 &\leqq 60 \end{aligned}}$$

$$\boxed{\begin{aligned} 2x_1 + 8x_2 + x_3 \qquad\quad &= 60 \\ 5x_1 + 2x_2 \qquad\quad + x_4 &= 60 \\ x_i \geqq 0 \qquad (i = 1, \cdots, 4). \end{aligned}}$$

Simplex Method

Maximize

$$z = f(x) = c_1 x_1 + \cdots + c_n x_n$$

subject to the constraints

$$a_{11} x_1 + \cdots + a_{1n} x_n = b_1$$

$$a_{21} x_1 + \cdots + a_{2n} x_n = b_2$$

$$\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

$$a_{m1} x_1 + \cdots + a_{mn} x_n = b_m$$

$$x_i \geqq 0 \qquad (i = 1, \cdots, n).$$

Simplex Method- Example

- The maximization of the objective function

$$z = 40x_1 + 88x_2$$

subject to

$$2x_1 + 8x_2 \leqq 60$$

$$5x_1 + 2x_2 \leqq 60$$

$$x_1 \geqq 0$$

$$x_2 \geqq 0.$$

Converting

$$\boxed{\begin{aligned} z - 40x_1 - 88x_2 \qquad\qquad &= 0 \\ 2x_1 + 8x_2 + x_3 \qquad &= 60 \\ 5x_1 + 2x_2 \qquad + x_4 &= 60 \end{aligned}}$$

Two slack variables x3 and x4

$$x_1 \geqq 0, \cdots, x_4 \geqq 0.$$

Simplex Method- Example

- Augmented matrix (simplex tableau or simplex table(the initial simplex table)):

$$
\begin{array}{c}
\begin{array}{cccccc} z & x_1 & x_2 & x_3 & x_4 & b \end{array} \\
\mathbf{T}_0 = \left[\begin{array}{c|cc|cc|c}
1 & -40 & -88 & 0 & 0 & 0 \\
\hline
0 & 2 & 8 & 1 & 0 & 60 \\
0 & 5 & 2 & 0 & 1 & 60
\end{array}\right]
\end{array}
$$

x_3 and x_4 are basic variables
and x_1 and x_2 are nonbasic variables

Simplex Method- Example

- by setting the nonbasic variables to zero, the basic feasible solution is obtained as

$$x_1 = 0, \qquad x_2 = 0,$$

$$x_3 = 60/1 = 60, \qquad x_4 = 60/1 = 60, \qquad z = 0$$

→The optimal solution (its location and value) is now obtained stepwise by pivoting, designed to take us to basic feasible solutions with higher and higher values of z until the maximum of z is reached

Simplex Method-Step 1.

- Operation O1: Selection of the Column of the Pivot: Select as the column of the pivot the first column with a negative entry in Row 1

First row

$$
(4) \qquad \mathbf{T}_0 = \begin{array}{c}
\begin{array}{cccccc} z & x_1 & x_2 & x_3 & x_4 & b \end{array} \\
\left[\begin{array}{c|cc|cc|c}
1 & \boxed{-40} & -88 & 0 & 0 & 0 \\
\hline
0 & 2 & 8 & 1 & 0 & 60 \\
0 & 5 & 2 & 0 & 1 & 60
\end{array}\right]
\end{array}
$$

In (4) this is Column 2 (because of the -40)

Simplex Method-Step 1.

- Operation O2: Selection of the Row of the Pivot: Divide the right sides [60 and 60 in (4)] by the corresponding entries of the column just selected ((60/2=30) > (60/5=12)) Take as the pivot equation the equation that gives the smallest quotient.

First row

$$
(4) \qquad \mathbf{T}_0 = \begin{array}{c} \begin{array}{cccccc} z & x_1 & x_2 & x_3 & x_4 & b \end{array} \\ \left[\begin{array}{c|cc|cc|c} 1 & \boxed{-40} & -88 & 0 & 0 & 0 \\ \hline 0 & 2 & 8 & 1 & 0 & 60 \\ 0 & \boxed{5} & 2 & 0 & 1 & 60 \end{array}\right] \end{array}
$$

Smallest

The pivot is 5 because is smallest.

Simplex Method-Step 1.

- Operation O3: Elimination by Row Operations. This gives zeros above and below the pivot

$$
(5) \qquad \mathbf{T}_1 = \begin{array}{c} \begin{array}{cccccc} z & x_1 & x_2 & x_3 & x_4 & b \end{array} \\ \left[\begin{array}{c|cc|cc|c} 1 & 0 & -72 & 0 & 8 & 480 \\ \hline 0 & 0 & 7.2 & 1 & -0.4 & 36 \\ 0 & \boxed{5} & 2 & 0 & 1 & 60 \end{array}\right] \end{array} \begin{array}{l} \text{Row 1} + 8 \text{ Row 3} \\ \text{Row 2} - 0.4 \text{ Row 3} \\ \\ \end{array}
$$

→basic variables are now x_1 and x_3 and nonbasic variables are x_2 and x_4

$$x_1 = 60/5 = 12, \qquad x_2 = 0,$$

$$\boxed{x_3 = 36/1 = 36, \qquad x_4 = 0, \qquad z = 480.}$$

Simplex Method-Step 2.

The basic feasible solution given by (5) is not yet optimal because of the negative entry -72 in Row 1. Accordingly, we perform the operations O1 to O3 again, choosing a pivot in the column of -72

$$
(5) \qquad \mathbf{T}_1 = \begin{array}{c} \begin{array}{cccccc} z & x_1 & x_2 & x_3 & x_4 & b \end{array} \\ \left[\begin{array}{c|cc|c|c|c} 1 & 0 & \boxed{-72} & 0 & 8 & 480 \\ \hline 0 & 0 & 7.2 & 1 & -0.4 & 36 \\ 0 & 5 & 2 & 0 & 1 & 60 \end{array}\right] \end{array} \begin{array}{l} \text{Row 1} + 8 \text{ Row 3} \\ \text{Row 2} - 0.4 \text{ Row 3} \\ \\ \end{array}
$$

Simplex Method-Step 2.

$$
(5) \qquad \mathbf{T}_1 = \begin{array}{c} \begin{array}{cccccc} z & x_1 & x_2 & x_3 & x_4 & b \end{array} \\ \left[\begin{array}{c|cc|cc|c} 1 & 0 & \boxed{-72} & 0 & 8 & 480 \\ \hline 0 & 0 & \boxed{7.2} & 1 & -0.4 & 36 \\ 0 & 5 & 2 & 0 & 1 & 60 \end{array}\right] \end{array} \begin{array}{l} \text{Row 1} + 8 \text{ Row 3} \\ \text{Row 2} - 0.4 \text{ Row 3} \\ \\ \end{array}
$$

Operation O_1. Select Column 3 of $\mathbf{T}_1$ in (5) as the column of the pivot (because $-72 < 0$).

Operation O_2. We have $36/7.2 = 5$ and $60/2 = 30$. Select 7.2 as the pivot (because $5 < 30$).

Operation O_3. Elimination by row operations gives

$$
(6) \qquad \mathbf{T}_2 = \begin{array}{c} \begin{array}{cccccc} z & x_1 & x_2 & x_3 & x_4 & b \end{array} \\ \left[\begin{array}{c|cc|cc|c} 1 & 0 & 0 & 10 & 4 & 840 \\ \hline 0 & 0 & 7.2 & 1 & -0.4 & 36 \\ 0 & 5 & 0 & -\dfrac{1}{3.6} & \dfrac{1}{0.9} & 50 \end{array}\right] \end{array} \begin{array}{l} \text{Row 1} + 10 \text{ Row 2} \\ \\ \text{Row 3} - \dfrac{2}{7.2} \text{ Row 2} \end{array}
$$

Simplex Method-Step 2.

$$
(6) \qquad \mathbf{T}_2 = \begin{array}{c} \begin{array}{cccccc} z & x_1 & x_2 & x_3 & x_4 & b \end{array} \\ \left[\begin{array}{c|cc|cc|c} 1 & 0 & 0 & 10 & 4 & 840 \\ \hline 0 & 0 & 7.2 & 1 & -0.4 & 36 \\ 0 & 5 & 0 & -\dfrac{1}{3.6} & \dfrac{1}{0.9} & 50 \end{array}\right] \end{array} \begin{array}{l} \text{Row } 1 + 10 \text{ Row } 2 \\ \\ \text{Row } 3 - \dfrac{2}{7.2} \text{ Row } 2 \end{array}
$$

→now x_1 and x_2 are basic and x_3 and x_4 are nonbasic

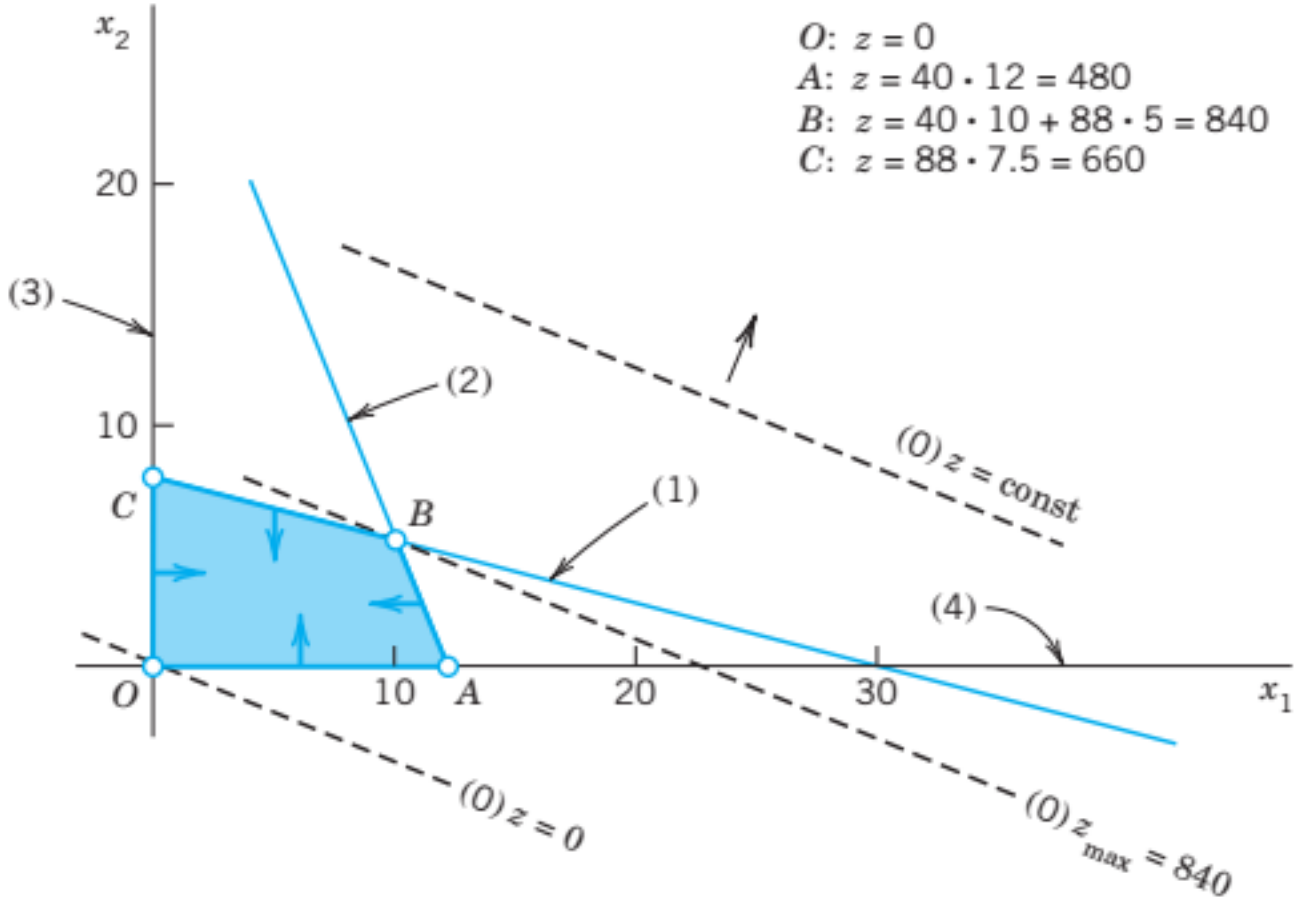
→the basic feasible solution

$$x_3 = 0, \qquad x_4 = 0,$$

$$x_1 = 50/5 = 10, \qquad x_2 = 36/7.2 = 5,$$

$$z = 840.$$

Since T_2 contains no more negative entries in Row 1→that z=f(10,5) = 40*10+88*5= 840 is the maximum possible revenue.



$$
\begin{array}{ll}
(0) & z = 40x_1 + 88x_2 \\
(1) & 2x_1 + 8x_2 \leqq 60 \\
(2) & 5x_1 + 2x_2 \leqq 60 \\
(3) & x_1 \geqq 0 \\
(4) & x_2 \geqq 0
\end{array}
$$

Exercises

Write in normal form and solve by the simplex method, assuming all x_J to be nonnegative

Maximize $f = 2x_1 + 3x_2 + x_3$ subject to
$x_1 + x_2 + x_3 \leqq 4.8,\ 10x_1 + x_3 \leqq 9.9,\ x_2 - x_3 \leqq 0.2.$

Solution

From the given data we obtain the augmented matrix

$$\mathbf{T}_0 = \begin{bmatrix} 1 & -2 & -3 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 4.8 \\ 0 & 10 & 0 & 1 & 0 & 1 & 0 & 9.9 \\ 0 & 0 & 1 & -1 & 0 & 0 & 1 & 0.2 \end{bmatrix}.$$

The pivot is 10. The indicated calculations give

$$\mathbf{T}_1 = \begin{bmatrix} 1 & 0 & -3 & -\frac{4}{5} & 0 & \frac{1}{5} & 0 & 1.98 \\ 0 & 0 & 1 & \frac{9}{10} & 1 & -\frac{1}{10} & 0 & 3.81 \\ 0 & 10 & 0 & 1 & 0 & 1 & 0 & 9.9 \\ 0 & 0 & 1 & -1 & 0 & 0 & 1 & 0.2 \end{bmatrix} \quad \begin{array}{l} \text{Row } 1 + \frac{2}{10} \text{ Row } 3 \\ \text{Row } 2 - \frac{1}{10} \text{ Row } 3 \\ \text{Row } 3 \\ \text{Row } 4 \end{array}$$

The next pivot is 1 in Row 4 and Column 3. The indicated calculations give

$$\mathbf{T}_2 = \begin{bmatrix} 1 & 0 & 0 & -\frac{19}{5} & 0 & \frac{1}{5} & 3 & 2.58 \\ 0 & 0 & 0 & \frac{19}{10} & 1 & -\frac{1}{10} & -1 & 3.61 \\ 0 & 10 & 0 & 1 & 0 & 1 & 0 & 9.9 \\ 0 & 0 & 1 & -1 & 0 & 0 & 1 & 0.2 \end{bmatrix} \quad \begin{matrix} \text{Row 1} + 3\ \text{Row 4} \\ \text{Row 2} - \text{Row 4} \\ \text{Row 3} \\ \text{Row 4} \end{matrix}$$

The last pivot needed is 19/10 in Row 2 and Column 4. We obtain

$$\mathbf{T}_3 = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 & 0 & 1 & 9.8 \\ 0 & 0 & 0 & \frac{19}{10} & 1 & -\frac{1}{10} & -1 & 3.61 \\ 0 & 10 & 0 & 0 & -\frac{10}{19} & \frac{20}{19} & \frac{10}{19} & 8 \\ 0 & 0 & 1 & 0 & \frac{10}{19} & -\frac{1}{19} & \frac{9}{19} & 2.1 \end{bmatrix} \quad \begin{matrix} \text{Row 1} + 2\ \text{Row 2} \\ \text{Row 2} \\ \text{Row 3} - \frac{10}{19}\ \text{Row 2} \\ \text{Row 4} + \frac{10}{19}\ \text{Row 2} \end{matrix}$$

Hence a solution of our problem is

$$f\left(\frac{8}{10}, \frac{2.1}{1}, \frac{3.61}{19/10}\right) = f(0.8, 2.1, 1.9) = 9.8.$$

Exercises:

Write in normal form and solve by the simplex method, assuming all x_J to be nonnegative

. Minimize $f = 4x_1 - 10x_2 - 20x_3$ subject to
$3x_1 + 4x_2 + 5x_3 \leqq 60, \quad 2x_1 + x_2 \leqq 20,$
$2x_1 + 3x_3 \leqq 30.$

Solution

$$\mathbf{T}_0 = \begin{bmatrix} 1 & -4 & 10 & 20 & 0 & 0 & 0 & 0 \\ 0 & 3 & 4 & 5 & 1 & 0 & 0 & 60 \\ 0 & 2 & 1 & 0 & 0 & 1 & 0 & 20 \\ 0 & 2 & 0 & 3 & 0 & 0 & 1 & 30 \end{bmatrix}$$

$$\mathbf{T}_1 = \begin{bmatrix} 1 & -\frac{23}{2} & 0 & \frac{15}{2} & -\frac{5}{2} & 0 & 0 & -150 \\ 0 & 3 & 4 & 5 & 1 & 0 & 0 & 60 \\ 0 & \frac{5}{4} & 0 & -\frac{5}{4} & -\frac{1}{4} & 1 & 0 & 5 \\ 0 & 2 & 0 & 3 & 0 & 0 & 1 & 30 \end{bmatrix} \quad \begin{array}{l} \text{Row } 1 - \frac{10}{4}\text{ Row } 2 \\ \text{Row } 2 \\ \text{Row } 3 - \frac{1}{4}\text{ Row } 2 \\ \text{Row } 4 \end{array}$$

$$\mathbf{T}_2 = \begin{bmatrix} 1 & -\frac{33}{2} & 0 & 0 & -\frac{5}{2} & 0 & -\frac{5}{2} & -225 \\ 0 & -\frac{1}{3} & 4 & 0 & 1 & 0 & -\frac{5}{3} & 10 \\ 0 & \frac{25}{12} & 0 & 0 & -\frac{1}{4} & 1 & \frac{5}{12} & \frac{35}{2} \\ 0 & 2 & 0 & 3 & 0 & 0 & 1 & 30 \end{bmatrix} \quad \begin{array}{l} \text{Row } 1 - \frac{15}{6} \text{ Row } 4 \\ \text{Row } 2 - \frac{5}{3} \text{ Row } 4 \\ \text{Row } 3 + \frac{5}{12} \text{ Row } 4 \\ \text{Row } 4 \end{array}$$

$f_{\min} = -225$ at $x_1 = 0$, $x_2 = 10/4 = 2.5$, $x_3 = 30/3 = 10$.